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Key Points:

- Fog intensity time series are time-irreversible
- TKE leads fog intensity in formation, no clear lead-lag relation exists in the mature stage, and fog intensity leads TKE in dissipation
- In fog formation only, the strength of the fog series' irreversibility increases linearly with the strength of its causal linkage to TKE

Supporting Information:

Supporting Information may be found in the online version of this article.

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Directional Dynamics of Fog: Irreversibility and Causal Coupling With Turbulence

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Abstract Fog prediction remains challenging because the physical processes governing its life cycle evolve across time scales and do not follow reversible or stationary dynamics. Using high-frequency visibility observations from Sable Island, Canada, this study analyzes fog intensity and turbulent kinetic energy (TKE) for their time irreversibility and causal relations. Fog intensity exhibits temporal asymmetry in all stages, while TKE remains nearly reversible. The lead-lag structure between the two variables is stage dependent: TKE leads fog intensity during formation, the coupling becomes symmetric during the mature phase, and fog intensity leads TKE during dissipation. Notably, during fog formation, the strength of fog's intrinsic irreversibility increases linearly with the strength of its causal linkage to TKE, revealing that fog initiation is governed by a directional sequence of turbulence-moisture interactions. These findings demonstrate that fog is a non-equilibrium, time-asymmetric system, and that capturing its stage-dependent directionality is required for enhanced fog prediction.

Plain Language Summary Fog is difficult to forecast because it forms, matures, and dissipates through several different processes that do not behave the same way over time. Using new measurements from Sable Island, Canada, this study examines how changes in fog thickness relate to changes in the swirling motion or turbulence near the ground. Directionality in time is based on the conjecture that “cause” must occur before or at the same time as the “effect.” Using this conjecture, the analysis shows that fog development and decay have a preferred direction in time, and that the connection between fog and turbulence depends on the stage of the fog event. Before fog forms, changes in turbulence occur first. Once fog is established, the two evolve together with no clear leader. During fog dissipation, the sequence reverses so that changes in the fog occur before changes in turbulence. A key finding is that fog shows the strongest time-asymmetric behavior when turbulence has the strongest influence during formation. This finding implies that fog begins through a directional chain of physical processes rather than through random fluctuations. Recognizing this stage-dependent directionality can improve future fog and visibility predictions.

1. Introduction

Fog is a near-surface meteorological phenomenon consisting of a suspension of water droplets that reduce the horizontal visibility below 1 km (WMO, 1996). Fog forms in various regions under a wide range of atmospheric conditions, and has undisputed impacts on transportation logistics, human safety, and ecological processes. Despite its importance, fog remains one of the most challenging atmospheric phenomena to predict, owing to the complex, multi-scale processes that govern its formation, duration, and dissipation (Bergot & Koracin, 2021; Gultepe et al., 2007). These challenges are particularly pronounced in marine environments, where fog formation depends not only on radiative cooling but also on air-sea temperature contrasts, moisture advection, turbulent mixing, and multiple microphysical-radiative feedbacks. Compared to inland radiation fog, which is often driven primarily by nocturnal surface cooling, marine fog reflects a more coupled and dynamically evolving boundary layer system (Koračin & Dorman, 2017; Koračin et al., 2014). As a result, even high resolution weather models struggle to capture fog's formation and dissipation in real time (Hintz et al., 2024; Price et al., 2015; Steeneveld et al., 2015), underscoring gaps in fog dynamics and motivating new approaches.

Because fog exhibits strong intermittency, persistence, and threshold-like behavior (Cornejo-Bueno et al., 2020; Huang et al., 2023; Salcedo-Sanz et al., 2021), a stochastic framework—already successful in rainfall studies—offers a promising avenue for advancing fog prediction. Rainfall has long been treated as a stochastic

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process in geophysics, with well-developed statistical models to represent its intermittent “on-off” behavior and intensity fluctuations (Gaume et al., 2007; Peters & Neelin, 2006; Rigby & Porporato, 2010; Rodríguez-Iturbe & Eagleson, 1987). Recent analyses have shown that fog event sequences share notable similarities with rainfall, such as comparable power-law statistics in the duration of dry periods and event sizes, as well as similar spectral signatures in their on–off switching behavior (Räsänen et al., 2018). These parallels further motivate the development of stochastic models “tailored” to fog.

A defining property of precipitation is its temporal irreversibility (also called time asymmetry), which reflects the inherent asymmetry between the rapid onset and gradual decay of storms, as well as the directional nature of moisture convergence, condensation, and runoff. Irreversibility in rainfall has been recognized as a critical property to incorporate into stochastic models (Molini et al., 2010), with Müller et al. (2017) showing that neglecting pronounced temporal asymmetry in synthetic rainfall series can bias hydrological applications, leading to reduced overflow predictions in urban drainage.

Building on these outcomes, this study uses field data collected on Sable Island, Canada to investigate the temporal irreversibility in fog, an essential property to account for in the development of stochastic frameworks that capture fog’s non-equilibrium dynamics. In contrast to previous studies that have largely focused on inland or coastal fog (Cornejo-Bueno et al., 2020; Räsänen et al., 2018; Salcedo-Sanz et al., 2021), the present work examines marine fog, where additional dynamical complexity arises from air–sea interactions and enhanced turbulent mixing. The analysis also links fog irreversibility to turbulent kinetic energy (TKE)—a known driver of the fog life cycle.

2. Methods of Analysis

2.1. Data

The data sets analyzed were collected on Sable Island, Canada, located off the coast of Nova Scotia (43.9337°N, 59.9149°W), over the month of July 2022 as part of the Fog and Turbulence Interactions in the Marine Atmospheres (FATIMA)—Grand Banks (GB) field campaign (Fernando et al., 2025). Over the observation period, fog occurred about 25% of the time. Time series of visibility (*vis*) was collected by a forward scatter sensor (Vaisala FD70) at a rate of 4 measurements per minute and a height of 2.5 m. Visibility measurements are used here as a bulk proxy for fog intensity, reflecting the integrated optical extinction from the droplet population rather than size-resolved microphysical properties. Frequency components lower than 300^{−1} Hz in the fog time series were filtered out. To draw similarities with studies conducted on the intensity of rainfall (Molini et al., 2010), the intensity of fog is defined as

$$I_{\text{fog}} = \frac{1}{vis}, \quad (1)$$

such that lower visibility *vis* corresponds to a higher intensity of fog. The use of 1/*vis* as a surrogate variable for the intensity of fog is further motivated by parametrization studies that show $vis^{-1} \sim (N_d LWC)^\gamma$, where N_d is the droplet number concentration, *LWC* is the liquid water content, and γ is a constant around 0.5 (Gultepe et al., 2017).

Three-dimensional component-wise velocity data were collected at 40 Hz with a sonic anemometer (Campbell Scientific IRGASON) located at a height of 2 m on the Center Tower (43.9337°N, 60.0224°W), which is approximately 20 m east of the FD70. Corrections to account for sonic path averaging and flow distortion by the sonic transducers are applied (Horst et al., 2015; Horst & Oncley, 2006), and frequency components lower than 300^{−1} Hz were filtered out. Turbulent kinetic energy (*k*) was then calculated as

$$k = \frac{1}{2} \left[\overline{(u')^2} + \overline{(v')^2} + \overline{(w')^2} \right], \quad (2)$$

where u' , v' , and w' are the longitudinal, transverse, and vertical velocity fluctuations, respectively, and overbar denotes time averaging such that *k* is down-sampled to 1/15 Hz to match the sampling rate of *vis* for calculations of cross correlations.

Table 1

Observed Fog Events Divided Into the Three Stages (Formation, Duration, and Dissipation)

Event	IOP	Synoptic	Date	Formation	Duration	Dissipation
1	1	HP/MX	03/07/2022	01:35–10:24	10:24–11:38	11:38–11:55
2	2	CG/MX	07/07/2022	00:36–03:27	03:27–03:36	03:36–04:06
3	3	LP/ST	10/07/2022	05:18–05:43	05:43–06:45	06:45–07:14
4	3	LP/ST	10/07/2022	20:30–23:15	23:15–02 : 34 ⁺¹	02 : 34 ⁺¹ –02 : 52 ⁺¹
5	5	HP/MX	14/07/2022	01:07–02:51	02:51–09:43	09:43–10:16
6	6	HP/ST	15/07/2022	22:26–22:54	22:54–23:58	23:58–00 : 17 ⁺¹
7	7	HP/ST	18/07/2022	13:47–18:07	18:07–20:53	20:53–21:16
8	8	HP/ST	18/07/2022	21:17–04 : 14 ⁺¹	04 : 14 ⁺¹ –15 : 15 ⁺¹	15 : 15 ⁺¹ –15 : 40 ⁺¹
9	9	HP/ST	21/07/2022	20:29–22:52	22:52–13 : 43 ⁺¹	13 : 43 ⁺¹ –14 : 14 ⁺¹
10	10	HP/ST	23/07/2022	21:35–22:26	22:26–13 : 15 ⁺¹	13 : 15 ⁺¹ –14 : 02 ⁺¹
11	10	HP/ST	24/07/2022	20:38–21:07	21:07–04 : 37 ⁺¹	04 : 37 ⁺¹ –05 : 50 ⁺¹
12	10	HP/ST	25/07/2022	19:03–20:40	20:40–22:08	22:08–22:36
13	11	CG/MX	26/07/2022	23:19–02 : 28 ⁺¹	02 : 28 ⁺¹ –10 : 32 ⁺¹	10 : 32 ⁺¹ –10 : 45 ⁺¹
14	12	CG/ST	27/07/2022	22:39–01 : 29 ⁺¹	01 : 29 ⁺¹ –05 : 46 ⁺¹	05 : 46 ⁺¹ –12 : 48 ⁺¹
15	13	HP/MX	30/07/2022	03:21–04:08	04:08–11:43	11:43–12:22
16	14	LP/MX	31/07/2022	05:46–06:17	06:17–07:18	07:18–11:11
17	–	–	10/07/2022	10:24–10:31	10:31–12:47	12:47–13:28
18	–	–	10/07/2022	15:39–17:20	17:20–19:16	19:16–20:17
19	–	–	11/07/2022	21:32–03 : 29 ⁺¹	03 : 29 ⁺¹ –09 : 34 ⁺¹	09 : 34 ⁺¹ –12 : 14 ⁺¹
20	–	–	15/07/2022	14:33–15:32	15:32–17:22	17:22–17:56
21	–	–	16/07/2022	04:35–04:47	04:47–07:39	07:39–08:27
22	–	–	18/07/2022	03:38–03:44	03:44–05:06	05:06–05:36
23	–	–	18/07/2022	05:47–05:53	05:53–07:30	07:30–07:53
24	–	–	18/07/2022	07:53–09:30	09:30–12:25	12:25–13:47
25	–	–	22/07/2022	16:29–19:51	19:51–20:07	20:07–23:29
26	–	–	22/07/2022	23:30–23:42	23:42–07 : 56 ⁺¹	07 : 56 ⁺¹ –11 : 17 ⁺¹
27	–	–	24/07/2022	15:29–16:08	16:08–19:51	19:51–20:14
28	–	–	25/07/2022	05:59–06:59	06:59–13:52	13:52–16:31

Note. Events 1–16 occur during intensive observation periods (IOP) which provide the synoptic set-ups—high pressure (HP), low pressure (LP), mixing (MX), stable (ST), or CG (convergence zone). More details about each IOP can be found in Fernando et al. (2025). Events 17–28 correspond to other low-visibility events that occur outside of IOPs. All times are in UTC.

Twenty-eight fog events were selected and listed in Table 1 and highlighted in Figure 1 (top). The fog events span a wide range of event lengths (from approximately 2 to 34 hr) and formation mechanisms (with the various synoptic set-ups included in Table 1 when available). To examine how time irreversibility and its coupling with TKE evolve over the fog life cycle, each fog event was divided into formation, duration, and dissipation stages and analyzed separately. The formation stage encompasses fog onset and development, the mature stage covers the period of sustained fog (peak intensity and local stationary state), and the dissipation stage marks the clearing phase. Accordingly, the separation of these stages was determined based on inspection of the temporal evolution of I_{fog} , with formation identified as the period from when I_{fog} begins to rise until it stabilizes and plateaus, duration as the period where I_{fog} remains approximately constant, and dissipation as the period where I_{fog} begins to decrease until it reaches a minimum. To illustrate, Figure 1 (bottom) displays formation, duration, and dissipation stages for fog event 15 occurring on 30/07/2022.

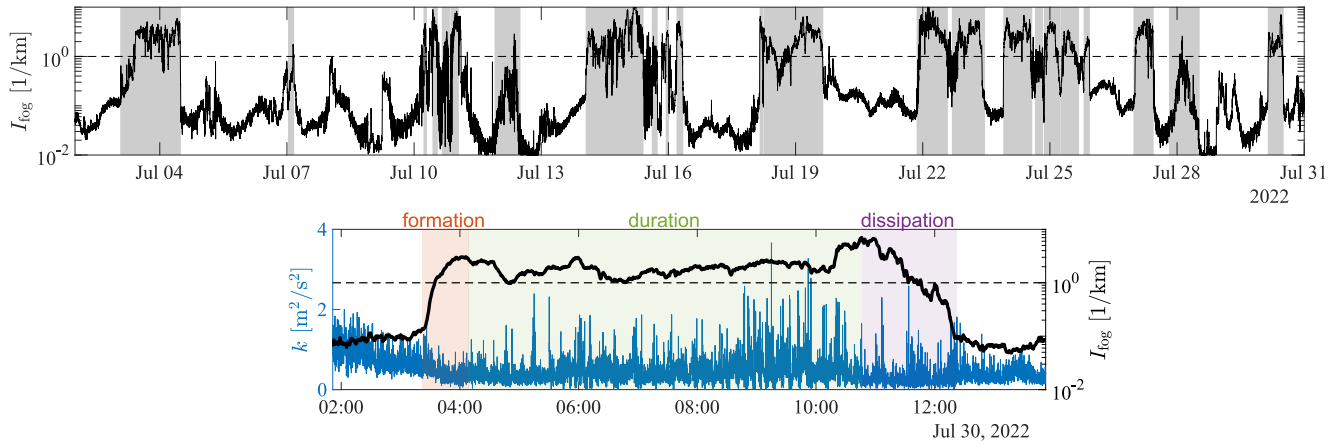


Figure 1. Top: Fog intensity for the duration of the FATIMA-GB campaign. Fog events selected are shaded in gray. The horizontal dashed line represents a fog intensity of 1 km^{-1} (or a visibility of 1 km), consistent with the threshold above which fog occurs. Bottom: Turbulent kinetic energy k and fog intensity during the fog event 15 on 30/07/2022 to illustrate the division of the event into formation (orange), duration (green), and dissipation (purple) stages. The horizontal dashed line represents a fog intensity of 1 km^{-1} (or a visibility of 1 km).

2.2. Time Irreversibility

A time series Θ is time reversible if the joint probability distribution of any sequence of its values is identical to the joint probability distribution of that same sequence taken in reverse order. A series that does not have this property is considered irreversible, or directional, and its statistics depend on the direction of time (Lawrance, 1991; Zorzetto et al., 2018). Here, the time irreversibility of both the fog intensity and TKE series is evaluated using three complementary metrics: (a) the lag irreversibility, (b) the Kullback–Leibler (KL) divergence between forward and reverse joint distributions, and (c) the cross-scale correlation asymmetry in wavelet space. These three measures are selected because they are based on different assumptions and simplifications so that their agreement implies robustness in the finding.

To be clear, rigorous evaluation of reversibility through the full n -point joint distributions is infeasible given the rapid growth in dimensionality. A weaker but more practical diagnostic is the so-called lag reversibility, which only requires bi-variate joint probability distributions

$$p_{\Theta_t, \Theta_{t+\tau}}(\Theta_0, \Theta_1) = p_{\Theta_{t+\tau}, \Theta_t}(\Theta_0, \Theta_1) \quad (3)$$

for all time lags $\tau = 1, 2, \dots$ and $t \in \mathbb{Z}$ (Lawrance, 1991). It then follows from lag reversibility that $\rho[\Theta_t^2, \Theta_{t+\tau}^2] = \rho[\Theta_t, \Theta_{t+\tau}^2]$, where $\rho[X, Y]$ denotes a correlation coefficient between variables X and Y . We define a lag irreversibility metric as (Zorzetto et al., 2018)

$$R(\tau) = \rho[\Theta_t^2, \Theta_{t+\tau}^2] - \rho[\Theta_t, \Theta_{t+\tau}^2] \quad (4)$$

such that $R(\tau) = 0$ for a perfectly reversible process and $|R(\tau)| > 0$ otherwise.

Another measure of irreversibility is the Kullback–Leibler divergence (or relative entropy) between the forward and reverse joint distributions. This metric is derived from the fluctuation theorem and involves conditional probabilities of a variable and its first time-derivative (Porporato et al., 2007):

$$\langle Z_\tau \rangle = \int p_\Theta(\Theta) \int p_\Theta(\dot{\Theta}|\Theta) \log \frac{p_\Theta(\dot{\Theta}|\Theta)}{p_\Theta(-\dot{\Theta}|\Theta)} d\Theta d\dot{\Theta}, \quad (5)$$

where the integrals extend over the whole domains of Θ and its time derivative $\dot{\Theta}$, which is approximated as $\Delta\Theta/\tau$. Similar to $|R(\tau)|$, $\langle Z_\tau \rangle = 0$ for a perfectly reversible process and $\langle Z_\tau \rangle > 0$ otherwise. In contrast to $R(\tau)$, which reflects asymmetry in the temporal ordering of values, $\langle Z_\tau \rangle$ incorporates the time derivative and therefore captures asymmetry in the rates of change of the time series. Together, these metrics provide complementary views of time irreversibility by capturing both value-based and rate-based asymmetries.

A third approach evaluated multiscale time directionality using a continuous wavelet transform (CWT) following Molini et al. (2010). The CWT decomposes the time series into scale-dependent coefficients from which local scale energies are derived. Directionality is then inferred from the asymmetry of the time-lagged cross-correlation between energies at adjacent scales: forward cascades yield stronger correlations at positive lags, inverse cascades at negative lags, and instantaneous processes produce symmetric correlations. This metric therefore provides a scale-dependent view of time asymmetry by assessing whether variability propagates across scales in a preferred temporal direction. However, as discussed later in Section 3, this approach is sensitive to the choice of wavelet basis and introduces smoothing through interpolation (continuous wavelet transforms produce high redundancy), so it is used only as a supplementary diagnostic.

2.3. Lead-Lag Relationships Between Fog Intensity and TKE

Moving beyond the irreversibility in a single time series, the lead-lag relation between fog intensity and TKE is examined through the asymmetry of their cross-correlation, $\rho[I_{\text{fog}}, k]$. To preserve the low-frequency variability that is physically relevant to fog-turbulence coupling, cross-correlations were computed using the raw (non-detrended) I_f and k series. For any pair of variables X and Y , the cross-correlation is defined as $\rho[X, Y] = \text{Corr}(X(t + \tau), Y(t))$. While the peak magnitude reflects the coupling strength, the peak offset and resulting asymmetry encode causal directionality. A peak at positive τ indicates that $Y(t)$ correlates most strongly with future $X(t + \tau)$, implying that Y precedes X . Conversely, a peak at negative τ implies that X precedes Y . A symmetric $\rho(\tau)$ (peaking near $\tau \approx 0$) suggests no preferred lead-lag relation. The degree of this asymmetry can be quantified as:

$$A = \int \frac{\rho(\tau) - \rho(-\tau)}{\rho(\tau) + \rho(-\tau)} d\tau, \quad (6)$$

where $\rho(\tau)$ and $\rho(-\tau)$ represent the cross-correlation at positive and negative lags, respectively. The measure A is the integrated form of Q in Jachens et al. (2006). Cases where changes in TKE lead variations in fog intensity yield $\rho(\tau) > \rho(-\tau)$ and thus $A > 0$, whereas the opposite yields $A < 0$. The scalar metric A therefore summarizes the full lead-lag dynamics into a single value, enabling direct comparison of the dominant causal direction across fog stages.

A more formal framework for interpreting this asymmetry is provided by linear response theory, which is used to describe the input-output properties of a system (Kubo, 1957). In this context, one variable is treated as an external perturbation and the other as the system's response. The asymmetry in their cross-correlation function then reveals the causal link, as a response cannot precede the action that causes it. The central tool for this analysis is the so-called susceptibility, $\chi(\omega)$, which is the Fourier transform of the cross-correlation function:

$$\chi(\omega) = \int_{-\infty}^{\infty} \rho(\tau) e^{-i\omega\tau} d\tau \quad (7)$$

The susceptibility is a complex-valued function where the real part, $\text{Re}\chi(\omega)$, corresponds to the symmetric (even) part of the correlation, and the imaginary part, $\text{Im}\chi(\omega)$, corresponds to the asymmetric (odd) part (Borysov & Balatsky, 2014). Since causality is an inherently asymmetric, time-ordered concept, the imaginary part becomes the primary indicator for analysis.

The sign of the peak in $\text{Im}\chi(\omega)$ for positive frequencies ($\omega > 0$) indicates the direction of the causal influence. Following the convention from applications in similar complex systems, a negative peak suggests that TKE acts as the input and fog intensity as the response (TKE $\rightarrow$ Fog). A positive peak suggests the reverse relation where fog intensity influences TKE (Fog $\rightarrow$ TKE). An imaginary part near zero would imply a symmetric relation with no clear lead-lag dynamic.

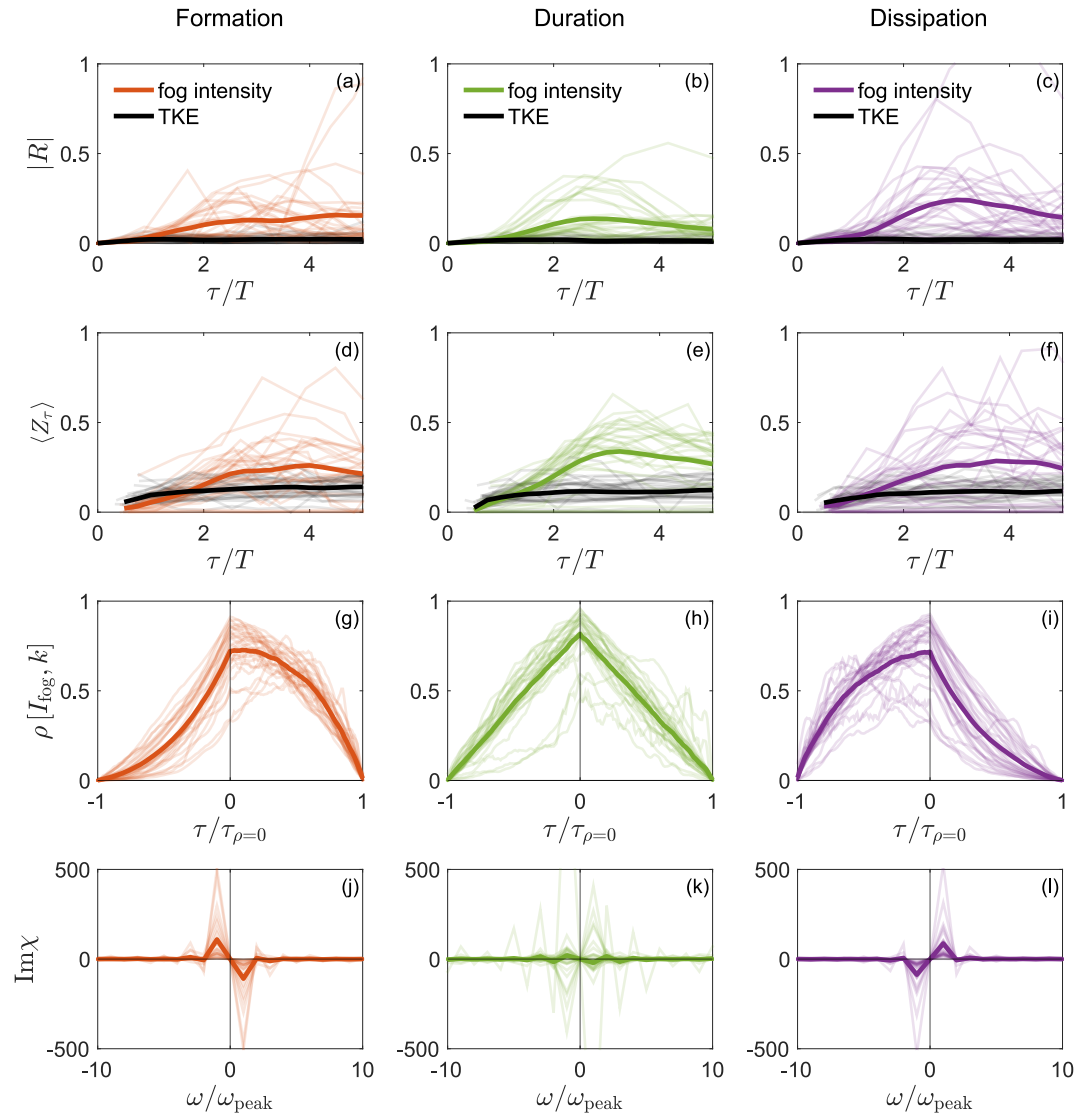


Figure 2. Lag irreversibility metric $|R|$ (a–c) and the Kullback–Leibler divergence $\langle Z_\tau \rangle$ (d–f) as functions of normalized lag τ/T , where T is the integral time scale. Cross-correlation between fog intensity and TKE $\rho[I_{\text{fog}}, k]$, with the time lag τ normalized by the time lag at which ρ crosses 0 to highlight the asymmetry (g–i). Imaginary component of the corresponding susceptibility $\text{Im}\chi$, with the frequency ω normalized by the frequency at which the positive peak occurs (j–l). The left, middle, and right columns correspond to the formation, mature, and dissipation stages of a fog event. For all figures, dark, thick lines show the metric evolution averaged across all fog events, with individual events plotted as lighter, thin lines.

3. Results

3.1. Time Irreversibility

The lag irreversibility metric $|R|$ and the Kullback–Leibler divergence $\langle Z_\tau \rangle$ are shown in Figures 2a–2c and 2d–2f, respectively, across the three life stages of fog. The absolute value of R is presented here to emphasize irreversibility magnitude, as its sign does not correspond to a physically interpretable direction. Fog intensity (colored lines) exhibits pronounced irreversibility, with both $|R|$ and $\langle Z_\tau \rangle$ significantly different from zero over a range of lags in all stages. In contrast, TKE (black lines) shows $|R| \approx 0$ and $\langle Z_\tau \rangle \approx 0$ across all stages, indicating approximately time-reversible statistics consistent with an instantaneous cascade. This contrast implies that the irreversibility of the fog intensity signal is not simply inherited from the turbulent kinetic energy but arises from additional radiative and microphysical processes governing fog evolution.

Time directionality is also assessed using the wavelet-based cross-scale representation, computed with the 4th-order Derivative of Gaussian (DoG) wavelet. Qualitatively, the fog intensity time series exhibits more temporal directionality than TKE in some events during the formation and dissipation stages, consistent with the results from the previous two metrics, although other events appear nearly symmetric. In the duration stage, most fog intensity cross-scale correlation functions are essentially symmetric, similar to that of TKE. The muted or symmetric wavelet-domain patterns likely result from the CWT's localization and redundancy, which smooth the signal and suppress low-frequency directional signatures. In the events that did exhibit a cascade direction, no consistent preference for forward or backward cascade direction is observed, with both directions occurring without an obvious connection to synoptic conditions. Moreover, the inferred cascade direction is sensitive to the choice of wavelet basis; in some cases a Morlet basis wavelet yields no or opposite signs of the asymmetry compared to the DoG basis wavelet. For these reasons, the time-domain metrics R and $\langle Z_\tau \rangle$ are treated as the primary indicators of irreversibility, and wavelet-based figures are provided in Supporting Information S1.

3.2. Lead-Lag Relations Between Fog Intensity and TKE

The cross-correlations between fog intensity and TKE are presented in Figures 2g–2i for each event and stage. During the formation stage, a consistent positive asymmetry ($A > 0$) is observed across the fog events. This indicates that changes in k lead changes in fog intensity. This result is consistent with the physical understanding that a period of weakening turbulence is a necessary precondition for fog to develop, as calmer conditions allow for the local accumulation of moisture. For the duration stage, the relation becomes largely symmetric. This suggests a contemporaneous coupling, where the established fog layer and the turbulence within it are in a state of quasi-equilibrium rather than a causal lead-lag dynamic (Dione et al., 2023). The causal direction reverses during the dissipation stage, which is marked by a clear negative asymmetry ($A < 0$), so that changes in fog intensity lead changes in TKE. Fog dissipation is often initiated by larger-scale phenomena such as changes in radiation or moisture availability. The subsequent breaking of the fog layer alters the stability of the atmosphere, which then enables new buoyancy-driven generation mechanisms that cause TKE to increase in response to the fog clearing (Dione et al., 2023).

Similar outcomes are seen in $\text{Im}\chi(\omega)$, Figures 2j–2l, which mirrors the stage-dependent asymmetry observed in the time-domain cross-correlations across all stages. The extrema in $\text{Im}\chi(\omega)$ occur at frequencies corresponding to timescales commensurate with the stage duration, reflecting the integration of asymmetry over the full lag range; thus, ω is normalized by the frequency of the dominant extremum (ω_{peak}) to emphasize the spectral shape. Agreement between the time-domain cross-correlation and its spectral analogue is not unexpected given the Wiener–Khinchin theorem; however, such convergence formally requires approximate stationarity. Thus, the consistency between the two domains may be interpreted as indirect evidence that non-stationarity did not play a dominant role in this analysis.

3.3. Time Irreversibility and Causal Coupling With TKE

A notable relation emerges when the intrinsic time irreversibility of the fog intensity signal is compared with the strength of its coupling to TKE. Figure 3 shows that during the formation stage, the lag irreversibility—integrated across five integral time scales to arrive at a single value for each fog event—increases linearly with the asymmetry measure A of $\rho[I_{\text{fog}}, k]$. Events in which TKE leads fog intensity (larger $A > 0$) also exhibit greater departure from time-reversible statistics. This coexistence of strong causal ordering and strong statistical irreversibility suggests that the physical processes governing fog initiation—particularly the rapid transition into a low-turbulence, moisture-accumulating regime—produce a pronounced temporal asymmetry in the fog evolution itself. This relation holds across all fog events, despite the wide range of synoptic setups and formation mechanisms (Table 1). No such relation appears during the duration or dissipation stages, indicating that the linkage between causal influence and intrinsic irreversibility is specific to the onset of fog formation. This interpretation is consistent with a threshold-based role for TKE at the onset of dissipation: rather than acting as a continuous driver, a brief spike in turbulence may initiate the breakup of the fog layer, but it does not subsequently control the rate or structure of the decay. In contrast to $R_{I_{\text{fog}}}$, the integrated $\langle Z_\tau \rangle_{I_{\text{fog}}}$, which reflects asymmetry in the joint distribution rather than directional cross-moment structure, does not exhibit the same linear trend with A , suggesting that it captures temporal asymmetry beyond what is represented by correlation-based statistics alone.

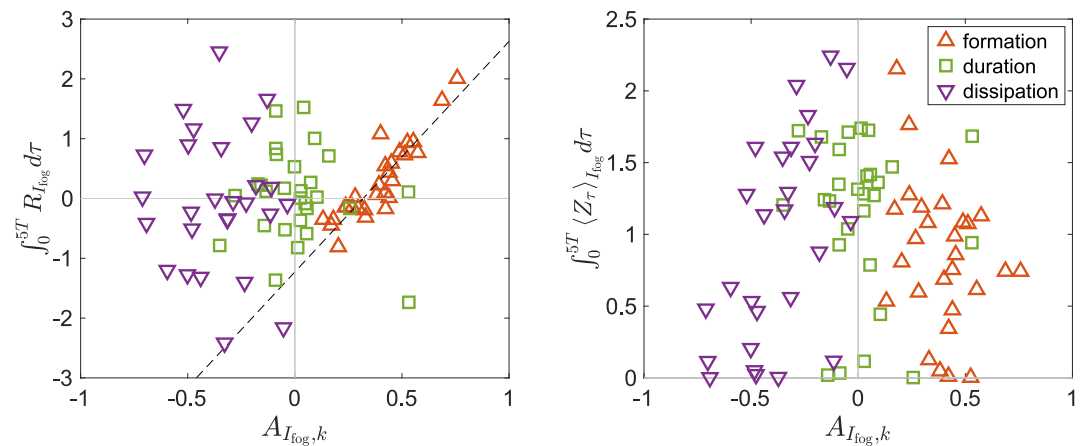


Figure 3. Time irreversibility of the fog time series based on R (left) and based on $\langle Z_\tau \rangle$ (right) versus the strength of TKE's causal influence (represented by A). To arrive at a single value representing the time irreversibility in each fog event, R and $\langle Z_\tau \rangle$ are integrated over $5T$, where T is the integral time scale. A linear relation, highlighted by the dashed line, emerges in the formation stage for the R metric.

Interestingly, corresponding analyses for no-fog reference cases show similar qualitative irreversibility metrics and stage-dependent asymmetry in the cross-correlation, but no comparable linear relation between $\int R_{I_{fog}} d\tau$ and A (Figures S5–S7 in Supporting Information S1). This suggests that while background boundary-layer dynamics generate asymmetry even without fog, the structured coupling between irreversibility and directional TKE influence emerges specifically during fog formation. Sensitivity to stage boundaries and window length is also assessed for the formation stage (Figures S8 and S9 in Supporting Information S1). Irreversibility and cross-correlation metrics are largely insensitive to boundary perturbations, while the $\int R_{I_{fog}} d\tau$ – A relationship is robust to the onset, strongest near the nominal end, and well resolved when $\geq 70\%$ of the formation stage is retained.

4. Conclusion and Discussion

The fog life cycle was analyzed through the statistical lenses of time irreversibility and causal inference by examining 28 fog events occurring on Sable Island, Canada, over the month of July 2022. The fog events spanned a wide range of synoptic conditions and formation mechanisms. Overall, fog intensity exhibits pronounced time irreversibility, whereas TKE remains nearly reversible across all stages of the fog life cycle. This contrast indicates that the directional structure in fog arises from thermodynamic and microphysical processes rather than from turbulence energetics alone. The lead–lag patterns between fog intensity and TKE further separate the stages: TKE leads fog intensity during formation, the coupling becomes largely symmetric during the mature stage, and fog intensity leads TKE during dissipation.

A key result is the emergence of a linear relation between fog's intrinsic irreversibility and the strength of its causal linkage to TKE only during fog formation. This alignment suggests that the onset of fog is governed by a directional sequence of turbulence–moisture interactions that imprint strong asymmetry on the fog signal. No such relation appears during duration or dissipation, consistent with quasi-equilibrium coevolution in the mature stage and with fog-driven stability changes dominating during decay. A brief turbulence spike may initiate breakup at the onset of dissipation, but it does not seem to govern the subsequent decay rate. Thus, variability during the duration and dissipation stages likely arises from threshold-like transitions that are not easily attributed to a single controlling variable. Although TKE captures key aspects of stability and surface forcing, resolving these effects more fully will require longer records and additional co-located measurements, providing a clear direction for future work.

These results extend earlier work showing that fog evolves through irregular bursts rather than smooth transitions, revealing that fog is both intermittent and time-asymmetric (Huang et al., 2023). Taken together, these findings establish a quantitative view of fog as a non-equilibrium, directionally biased system whose stages are governed by different physical processes. Capturing this stage-dependent asymmetry will likely lead to improvements in fog prediction.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The data set analyzed is available at Huang and Fernando (2026).

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